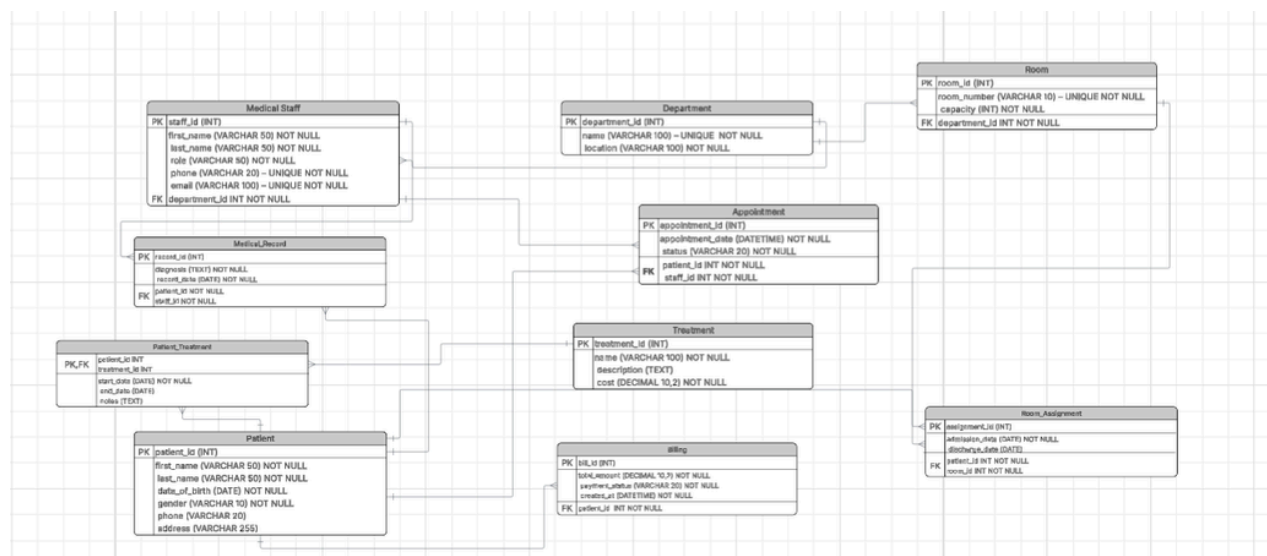


1.1 Logical Scheme



1.2 Objects

Table Descriptions

Medical_Staff

The Medical_Staff table stores information about the hospital staff such as doctors, nurses, and other employees.

It includes their names, roles, contact information, and the department where they work.

Department

The Department table stores information about hospital departments.

Each department has a unique identifier, a name, and a location.

Room

The Room table stores information about hospital rooms or wards.

It includes the room number, its capacity, and the department to which the room belongs.

Patient

The Patient table contains personal information about patients.

It includes their first name, last name, date of birth, gender, phone number, and address.

Appointment

The Appointment table stores information about patient appointments with medical staff.

It includes the appointment date and time, appointment status, and references to the patient and the staff member.

Medical_Record

The Medical_Record table stores medical records of patients.
It contains the diagnosis, the date of the record, and references to the patient and the staff member who created the record.

Treatment

The Treatment table contains information about different types of medical treatments or procedures.
It includes the treatment name, description, and cost.

Patient_Treatment

The Patient_Treatment table links patients with the treatments they receive.
It stores the start date, end date, and additional notes related to the treatment.

Room_Assignment

The Room_Assignment table stores information about patient room assignments.
It includes the admission date, discharge date, and references to the patient and the room.

Billing

The Billing table stores information about patient bills.
It includes the total amount, payment status, bill creation date, and the patient identifier.

Table: Department

Field Name	Field Description	Data Type / Constraints
department_id	Unique department identifier	INT, PK
name	Department name	VARCHAR(100), UNIQUE, NOT NULL
location	Department location	VARCHAR(100), NOT NULL

Table: Medical_Staff

Field Name	Field Description	Data Type / Constraints
staff_id	Unique staff identifier	INT, PK
first_name	Staff first name	VARCHAR(50), NOT NULL
last_name	Staff last name	VARCHAR(50), NOT NULL
role	Staff position	VARCHAR(50), NOT NULL
phone	Staff phone number	VARCHAR(20), UNIQUE
email	Staff email	VARCHAR(100), UNIQUE
department_id	Department of staff	INT, FK

Table: Patient

Field Name	Field Description	Data Type / Constraints
patient_id	Unique patient identifier	INT, PK
first_name	Patient first name	VARCHAR(50), NOT NULL
last_name	Patient last name	VARCHAR(50), NOT NULL
date_of_birth	Date of birth	DATE, NOT NULL
gender	Patient gender	VARCHAR(10)
phone	Patient phone	VARCHAR(20)
address	Patient address	VARCHAR(255)

Table: Appointment

Field Name	Field Description	Data Type / Constraints
appointment_id	Unique appointment identifier	INT, PK
appointment_date	Date and time of appointment	DATETIME, NOT NULL
status	Appointment status	VARCHAR(20), NOT NULL
patient_id	Patient identifier	INT, FK
staff_id	Doctor identifier	INT, FK

Table: Medical_Record

Field Name	Field Description	Data Type / Constraints
record_id	Medical record ID	INT, PK
diagnosis	Patient diagnosis	TEXT
record_date	Date of record	DATE
patient_id	Patient identifier	INT, FK
staff_id	Staff identifier	INT, FK

Table: Treatment

Field Name	Field Description	Data Type / Constraints
treatment_id	Treatment identifier	INT, PK
name	Treatment name	VARCHAR(100)
description	Treatment description	TEXT
cost	Treatment cost	DECIMAL(10,2)

Table: Patient_Treatment

Field Name	Field Description	Data Type / Constraints
patient_id	Patient identifier	INT, PK, FK
treatment_id	Treatment identifier	INT, PK, FK
start_date	Treatment start date	DATE
end_date	Treatment end date	DATE
notes	Additional notes	TEXT

Table: Room

Field Name	Field Description	Data Type / Constraints
room_id	Room identifier	INT, PK
room_number	Room number	VARCHAR(10), UNIQUE
capacity	Number of beds	INT
department_id	Department identifier	INT, FK

Table: Room_Assignment

Field Name	Field Description	Data Type / Constraints
assignment_id	Assignment identifier	INT, PK
admission_date	Admission date	DATE
discharge_date	Discharge date	DATE
patient_id	Patient identifier	INT, FK
room_id	Room identifier	INT, FK

Table: Billing

Field Name	Field Description	Data Type / Constraints
bill_id	Billing identifier	INT, PK
total_amount	Total bill amount	DECIMAL(10,2)
payment_status	Payment status	VARCHAR(20)
bill_date	Date of bill	DATETIME
patient_id	Patient identifier	INT, FK

Comments on table relationships

- Department → Medical Staff : 1 : Many
- Patient → Appointment : 1 : Many
- Medical Staff → Appointment : 1 : Many
- Patient → Medical Record : 1 : Many
- Patient ↔ Treatment : Many : Many (by table Patient_Treatment)
- Department → Room : 1 : Many
- Patient → Room_Assignment : 1 : Many

Example with data

Patient

patient_id	first_name	last_name	date_of_birt h	gender	phone	address
1	John	Smith	1990-05-12	Male	87011234567	Almaty
2	Anna	Brown	1985-11-23	Female	87019876543	Astana
3	David	Lee	1992-07-08	Male	87015551234	Atyrau
4	Maria	Garcia	1988-03-19	Female	87017771234	Shymkent
5	Alex	Johnson	1995-01-10	Male	87014445678	Aktobe
6	Sofia	Wilson	1993-09-14	Female	87019998877	Karaganda

Medical_Staff

staff_id	first_name	last_name	role	phone	email	department_id
1	Michael	Adams	Doctor	87012223344	adams@hospital.com	1
2	Sarah	Miller	Nurse	87013334455	miller@hospital.com	1
3	Robert	Davis	Surgeon	87014445566	davis@hospital.com	2
4	Emily	Clark	Doctor	87015556677	clark@hospital.com	3

Department

department_id	name	location
1	Cardiology	Floor 1
2	Surgery	Floor 2
3	Neurology	Floor 3
4	Pediatrics	Floor 4

Room

room_id	room_number	capacity	department_id
1	101	2	1
2	102	3	1
3	201	2	2
4	301	1	3

Appointment

appointment_id	appointment_date	status	patient_id	staff_id
1	2025-05-10 10:00	Completed	1	1
2	2025-05-11 11:30	Scheduled	2	4
3	2025-05-12 09:00	Completed	3	1
4	2025-05-13 14:00	Cancelled	4	3
5	2025-05-14 15:30	Scheduled	5	2

Treatment

treatment_id	name	description	cost
1	Heart Check	Heart examination	150.00
2	Surgery	Surgical procedure	1200.00
3	MRI Scan	Brain MRI scan	500.00
4	Physical Therapy	Rehabilitation treatment	200.00

Patient_Treatment

patient_id	treatment_id	start_date	end_date	notes
1	1	2025-05-10	2025-05-10	Routine check
2	3	2025-05-11	2025-05-11	MRI scan
3	2	2025-05-12	2025-05-20	Surgery recovery
4	4	2025-05-13	2025-06-01	Therapy sessions

Room_Assignment

assignment_id	admission_date	discharge_date	patient_id	room_id
1	2025-05-10	2025-05-12	1	1
2	2025-05-11	2025-05-15	2	2
3	2025-05-12	2025-05-18	3	3
4	2025-05-13	2025-05-20	4	4

Billing

bill_id	total_amount	payment_status	bill_date	patient_id
1	150.00	Paid	2025-05-10	1
2	500.00	Paid	2025-05-11	2
3	1200.00	Pending	2025-05-12	3
4	200.00	Paid	2025-05-13	4